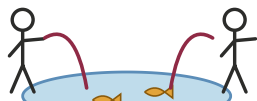


Extractive dilemma

Overuse of a shared resource depletes it.

HUMANS

don't overfish!



LLM AGENTS



shared budget



In humans

overfishing, overgrazing

In LLM agents

API budgets, shared compute, tool calls, context windows

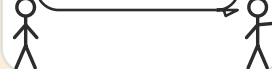
Contributive dilemma

Beneficiaries fail to maintain community goods.

HUMANS



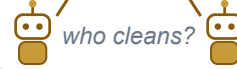
cite sources!



LLM AGENTS



who cleans?



In humans

open-source upkeep, unpaid moderation

In LLM agents

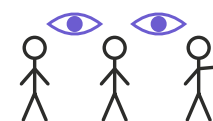
helping at a personal cost, cleanup, verifying, volunteering

Second-order dilemma

Monitoring is itself costly to provide.

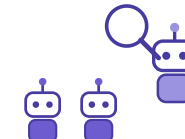
HUMANS

who watches you?



LLM AGENTS

monitor costs



In humans

peer monitoring, costly security, punishment

In LLM agents

auditing, sanctioning agents, setting new norms

Repair dilemma

Who bears the cost of fixing harm?

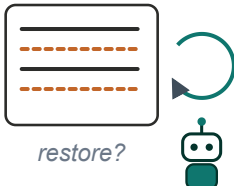
HUMANS

not me!

not me!



LLM AGENTS



restore?

In humans

restitution, policy reform, disaster recovery

In LLM agents

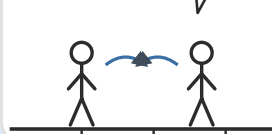
cleaning corrupted memory, retractions, undoing changes, restoring what was lost

Coordination dilemma

Several acceptable options; the group must pick one.

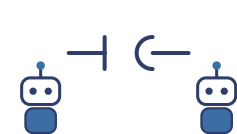
HUMANS

excuse me!



LLM AGENTS

v1 or v2?



In humans

which convention or standard to adopt

In LLM agents

which protocol or format to share

Collusion

Agents cooperate all too effectively.

HUMANS

psst...



LLM AGENTS

hidden channel



In humans

monitors and monitored collude; captured mediators

In LLM agents

agents coordinating to evade human oversight



shared-resource dilemmas



beyond the shared-resource framing